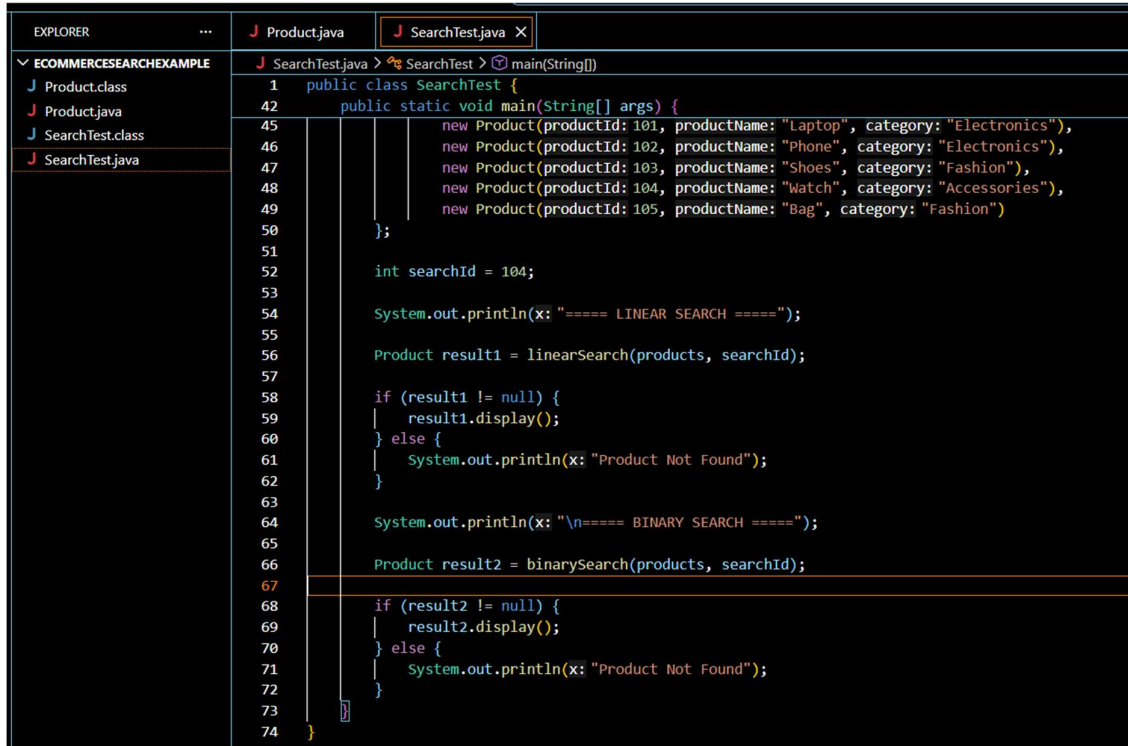
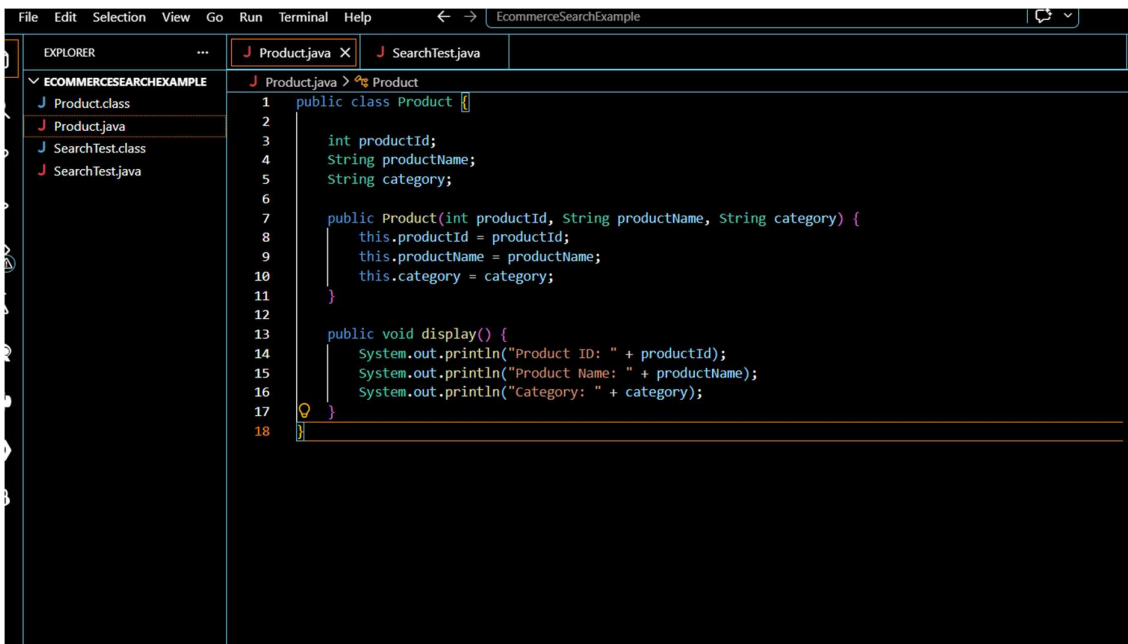


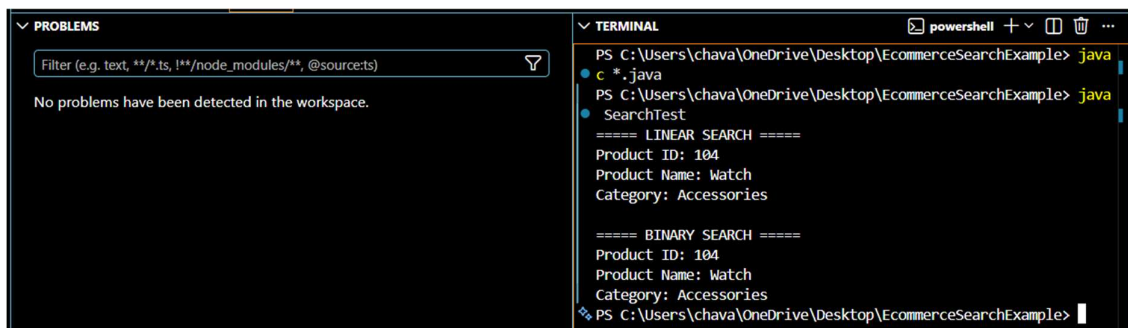
Exercise 2: E-commerce Platform Search Function



```
1 public class SearchTest {
42 public static void main(String[] args) {
45     new Product(productId: 101, productName: "Laptop", category: "Electronics"),
46     new Product(productId: 102, productName: "Phone", category: "Electronics"),
47     new Product(productId: 103, productName: "Shoes", category: "Fashion"),
48     new Product(productId: 104, productName: "Watch", category: "Accessories"),
49     new Product(productId: 105, productName: "Bag", category: "Fashion")
50 };
51
52 int searchId = 104;
53
54 System.out.println(x: "==== LINEAR SEARCH =====");
55
56 Product result1 = linearSearch(products, searchId);
57
58 if (result1 != null) {
59     result1.display();
60 } else {
61     System.out.println(x: "Product Not Found");
62 }
63
64 System.out.println(x: "\n==== BINARY SEARCH =====");
65
66 Product result2 = binarySearch(products, searchId);
67
68 if (result2 != null) {
69     result2.display();
70 } else {
71     System.out.println(x: "Product Not Found");
72 }
73
74 }
```



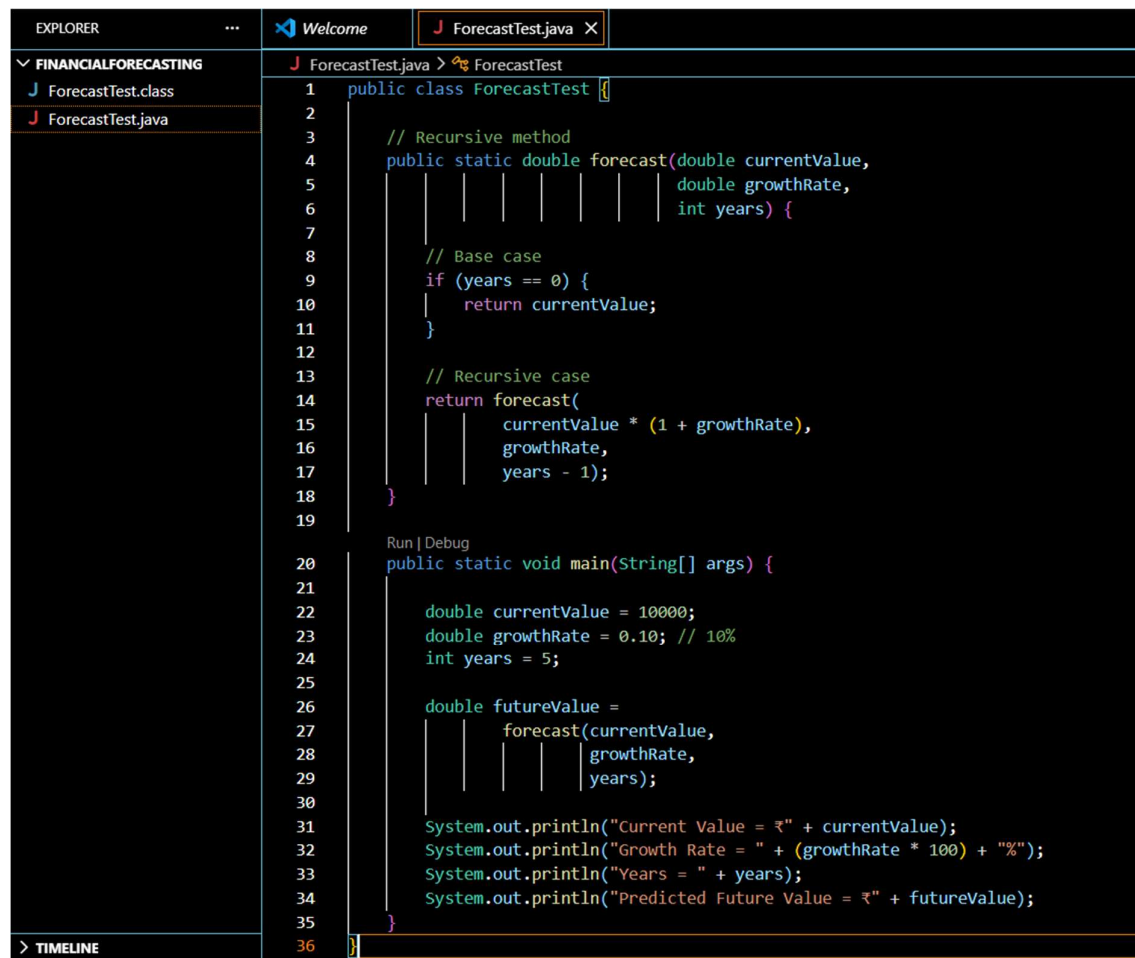
```
1 public class Product {
2
3     int productId;
4     String productName;
5     String category;
6
7     public Product(int productId, String productName, String category) {
8         this.productId = productId;
9         this.productName = productName;
10        this.category = category;
11    }
12
13    public void display() {
14        System.out.println("Product ID: " + productId);
15        System.out.println("Product Name: " + productName);
16        System.out.println("Category: " + category);
17    }
18 }
```



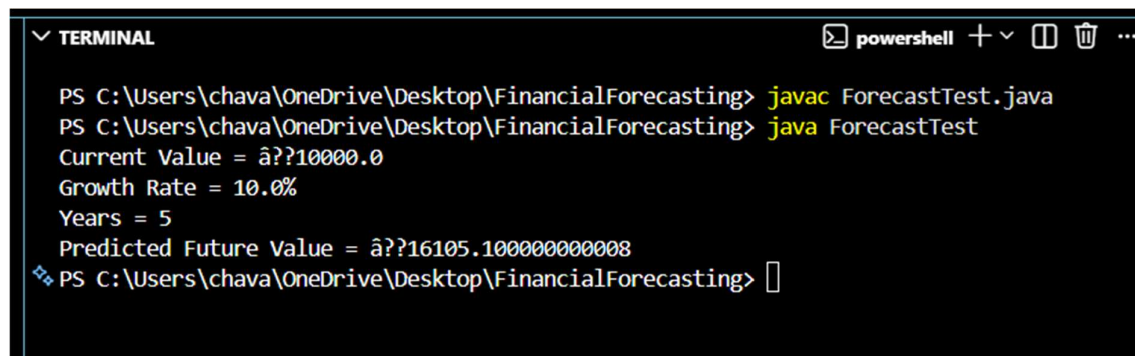
```
PS C:\Users\chava\OneDrive\Desktop\EcommerceSearchExample> java
c *.java
PS C:\Users\chava\OneDrive\Desktop\EcommerceSearchExample> java
SearchTest
==== LINEAR SEARCH =====
Product ID: 104
Product Name: Watch
Category: Accessories

==== BINARY SEARCH =====
Product ID: 104
Product Name: Watch
Category: Accessories
PS C:\Users\chava\OneDrive\Desktop\EcommerceSearchExample>
```

Exercise 7: Financial Forecasting



```
1 public class ForecastTest {
2
3     // Recursive method
4     public static double forecast(double currentValue,
5                                   double growthRate,
6                                   int years) {
7
8         // Base case
9         if (years == 0) {
10             return currentValue;
11         }
12
13         // Recursive case
14         return forecast(
15             currentValue * (1 + growthRate),
16             growthRate,
17             years - 1);
18     }
19
20     Run | Debug
21     public static void main(String[] args) {
22
23         double currentValue = 10000;
24         double growthRate = 0.10; // 10%
25         int years = 5;
26
27         double futureValue =
28             forecast(currentValue,
29                     growthRate,
30                     years);
31
32         System.out.println("Current Value = ₹" + currentValue);
33         System.out.println("Growth Rate = " + (growthRate * 100) + "%");
34         System.out.println("Years = " + years);
35         System.out.println("Predicted Future Value = ₹" + futureValue);
36     }
37 }
```



```
PS C:\Users\chava\OneDrive\Desktop\FinancialForecasting> javac ForecastTest.java
PS C:\Users\chava\OneDrive\Desktop\FinancialForecasting> java ForecastTest
Current Value = ₹10000.0
Growth Rate = 10.0%
Years = 5
Predicted Future Value = ₹16105.100000000008
PS C:\Users\chava\OneDrive\Desktop\FinancialForecasting>
```